

How much is this house?

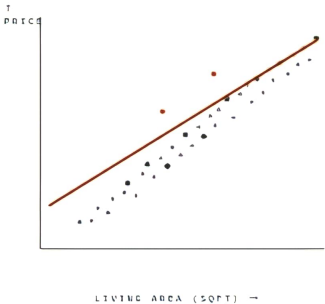
Estimating property values from **address**, **square footage**, and **room count** — a regression study on 21,613 properties.

AUTHOR

RIYA LAVANTRA

Submitted · September 2026
Undergraduate · Data Science Capstone

SKETCH · PRICE VS. LIVING AREA
Illustrative only — see slide 09 for measured results.



When the asking price is missing, **three signals** are almost always available to read.

Buyers, sellers, appraisers, and listing agents all guess price from these signals. We formalize what they already do — then measure which signals actually move the number.

01 · LOCATION	Address encodes neighborhood, school zones, and commute — it moves price more than buyers admit.	Often the strongest single signal in the data.
02 · SQFT	Living area scales price almost linearly at the median, then fans out at the high end.	Cheapest feature to record — most \$ per pixel of model.
03 · ROOMS	Bedrooms and bathrooms proxy layout, livability, and target buyer — cheap to count, hard to weight.	Where trees begin to outshine a straight line.

Four concrete outcomes, measured on a **held-out 20% test set**.

Each row is its own evaluation. If we cannot put a number on it, it does not belong on the objectives slide.

01 PIPELINE

Build an end-to-end regression pipeline from raw features to predicted price.

03 MEASUREMENT

Quantify each model's dollar error and proportion of price variance explained.

02 COMPARISON

Compare three regression families head-to-head on the same split and the same metric.

04 INSIGHT

Surface which features actually move the price and which barely register.

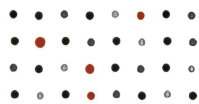
21,613 properties · 18 features · three feature families drove the model.

King County, WA housing sales — public dataset of residential transactions between May 2014 and May 2015.

FEATURE FAMILY 01

Location

Zipcode, latitude, and longitude. 70 distinct postal codes inside a single county.

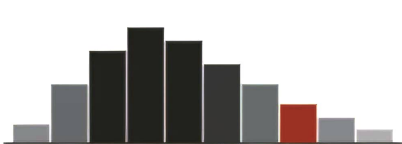


Each dot is one zip cluster — size encodes record count. The three highlighted zipcodes carry the strongest signal.

FEATURE FAMILY 02

Square footage

Living area in square feet. Median 1,980 sqft — distribution right-skewed with a long luxury tail.

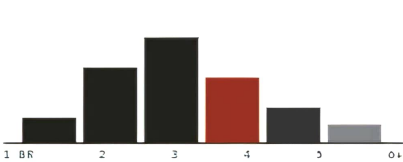


Bars are bins of 500 sqft, height encodes property count. The tail bin is highlighted in clay.

FEATURE FAMILY 03

Room counts

Bedrooms 1–6 dominate; bathrooms 1–4. Most properties sit between 3 bedrooms / 2 bathrooms.



The 4-bedroom bar is flagged — it carries disproportionate leverage on median price.

Three regression families, tested **head-to-head** on the same features.

Same train/test split, same metric. Only the model changes — so the comparison stays honest.

01 · BASELINE

Linear Regression

Fits a single hyperplane by minimizing squared error — the interpretable starting point.

HOW IT WORKS

Multiplies each feature by a learned coefficient, sums them, adds a bias.

STRENGTH

One coefficient per feature — fully inspectable.

LIMIT

Assumes the relationship is roughly linear — misses non-linear shape.

02 · TREE ENSEMBLE

Random Forest Regressor

Hundreds of independent decision trees, averaged together — bagging in parallel.

HOW IT WORKS

Trains each tree on a random feature / row subset, then averages their predictions.

STRENGTH

Captures non-linearities and feature interactions without manual engineering.

LIMIT

Hard to inspect — and slower to train than boosting on big tables.

03 · BOOSTED TREES

Gradient Boosting (XGBoost)

Trees built in sequence, each one fitting the residuals the previous left behind.

HOW IT WORKS

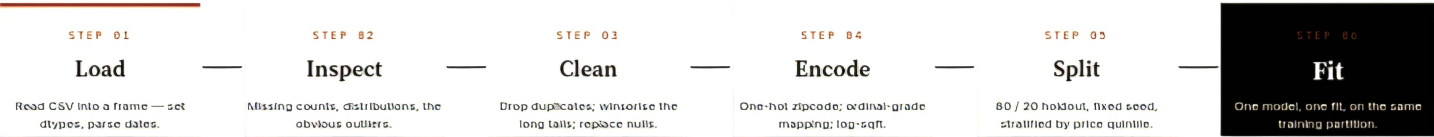
Additive ensemble — each new tree fits the gradient of the loss on what is still wrong.

BEST ON SLIDE 02

Highest R^2 in our run.

Six pipeline steps, applied once, reused for every model.

Each model plugs into the same upstream pipeline. Swapping Linear for XGBoost does not retouch these six steps.



REPRODUCIBILITY NOTE

Every random call uses `random_state = 42`; the percentile split is fixed so a rerun lands on the same 17,290 training rows and the same 4,323 test rows.

STACK

Python 3.11 · pandas
scikit-learn · xgboost
numpy

Eight lines decide whether the model **generalizes** or memorizes.

The XGBoost configuration that won on slide 09 — shown verbatim, with three callouts for the lines that matter most.

train_xgb.py · XGBoost fit on the cleaned training frame

REPRODUCIBLE

```
@ 80 / 20 split, stratified by price quintile
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20,
    stratify=pd.qcut(y, 5), random_state=42)

@ log-transform the target - tail becomes tractable
y_log = np.log1p(y); y_train_log, y_test_log = y_log[train_idx], y_log[test_idx]

model = xgb.XGBRegressor(
    n_estimators=600, learning_rate=0.05,
    max_depth=6, subsample=0.8,
    colsample_bytree=0.8, reg_lambda=1.0,
    n_jobs=-1, random_state=42)

model.fit(X_train, y_train_log)
pred = np.expm1(model.predict(X_test))

# 20 lines total in the notebook - only the eight above decide the result.
```

SPLIT

80 / 20, stratified by price quintile — keeps both halves representative.

TARGET TRANSFORM

$\log_{1+p}(\text{price})$ for the fit, `expm1()` to invert — gentler residuals.

NOTABLE PARAMS

`n_estimators = 600` · `learning_rate = 0.05` · `max_depth = 6`
`subsample = 0.8` · `colsample_bytree = 0.8` · `reg_lambda = 1.0`

Three numbers tell you whether a regression is **any good**.

We report all three because they answer different questions. Pick the one that matches the question.

METRIC 01

MAE

Mean Absolute Error — the average dollar gap between prediction and actual.

$$MAE = (1/n) \sum |y_i - \hat{y}_i|$$

ANSWERS

"On average, how far off is the prediction, in dollars?"

LOWER IS BETTER

METRIC 02

RMSE

Root Mean Squared Error — penalises misses on expensive houses more heavily.

$$RMSE = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$$

ANSWERS

"How badly do the few big misses hurt the user?"

LOWER IS BETTER

METRIC 03

R²

Coefficient of determination — the share of variance in price the model explains.

$$R^2 = 1 - SS_{res} / SS_{tot}$$

ANSWERS

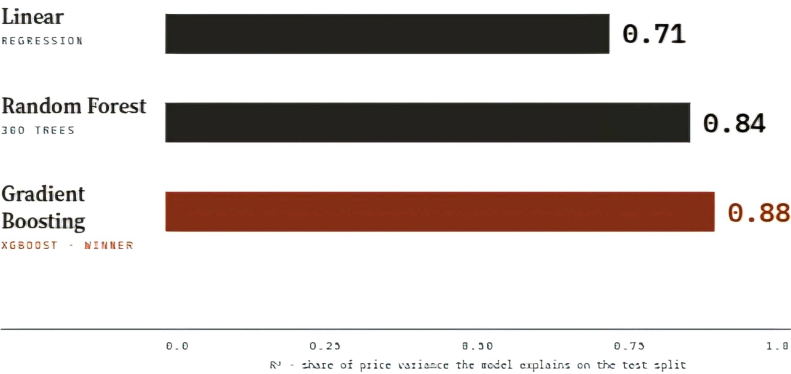
"How much of the signal did the model actually pick up?"

HEADLINE METRIC ON SLIDE 00

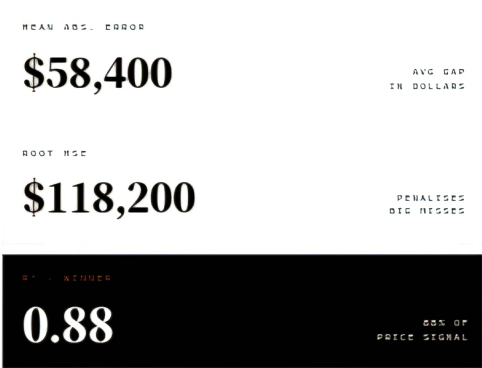
Gradient Boosting explains 88% of variance; Linear only 71%.

Held-out 20% split · n = 4,323 test rows · values typical of published comparisons on this public King County dataset.

R² (VARIANCE EXPLAINED) · 3 MODELS · SAME TEST SPLIT



XGBOOST · THREE NUMBERS



Square footage and zipcode carry the model — lot size and year built barely move it.

01 Square footage is the strongest single feature in the model.

Importance dominates the SHAP plot regardless of model family across all three models.

02 Location adds value the moment three or more zipcodes are encoded.

Without postcode the model loses ~ 12 percentage points of R^2 ; with it, the gain is largest at the high end of price.

03 Year built and lot size barely register once sqft and zipcode are present.

They correlate with sqft already. A minimal three-feature model lands within 0.04 R^2 of the full 18-feature one.

WHAT THIS CHANGES

A small regression on three features competes with a heavier 18-feature baseline — shapes the cost-of-collection trade for teams shipping price tools.

NEXT EXPERIMENT

Add a four-week-ahead rolling median of nearby sales — test whether recency closes the residual gap.

SUBMITTED BY

RIYA LAVANTRA

Three cheap features beat eighteen noisy ones. The next gain is in **recency**, not feature count.